IIIT HYDERABAD / VOICE AI ASSIGNMENT

FastSpeech 2 for English and Telugu Speech Synthesis

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Abstract

Two FastSpeech 2 models were trained separately: one on Svarah English and one on IndicTTS Telugu. Each used about five hours of training audio and one hour of validation audio. The work adapts the reference implementation to these datasets, prepares duration, pitch and energy targets, and generates speech with a pretrained HiFi-GAN vocoder. Both training runs finished, and the models generated all ten test utterances. The English outputs scored below the held-out baseline on the transcript-alignment check; Telugu scores were closer to their baseline. These measurements do not replace listening, which is still needed to assess pronunciation and prosody.

1. Introduction

The assignment is to train English and Telugu speech-synthesis models and examine how they handle the supplied test sentences. FastSpeech 2 [1] is useful for this task because it predicts duration, pitch and energy separately. Those predictions can be plotted alongside the generated audio to examine timing and prosody.

The starting code is ming024/FastSpeech2 [2]. The two models use the same code but are initialized and trained independently. Each has its own speaker embeddings, feature statistics and optimizer state. The pretrained vocoder is shared and kept fixed.

Experiment overview

Language	Training (hours)	Validation (hours)	Training run (steps)	Selected checkpoint (step)
English	5.0014	1.0020	19,000	7,000
Telugu	5.0001	1.0009	26,000	14,000

Synthesis pipeline



Each test utterance has a WAV file and separate duration, pitch and energy plots: ten audio files and thirty plots in total.

Code checks

The saved verification run passed all 12 tests, static checks and Python compilation. Tests cover text processing, data splits, duplicate removal, duration targets and model operations. A full forward and backward pass also completed on the local RTX 4050 GPU. These checks verify execution, not speech quality.

2. Model implementation

The text encoder maps input tokens to hidden features. A speaker embedding is added to these features before they enter the variance adaptor. The main settings are listed in Table 1.

Table 1. Acoustic-model configuration.

Component	Setting
Encoder and decoder	Four feed-forward Transformer blocks each; hidden size 192; two attention heads
Feed-forward convolution	768 channels; kernel sizes 9 and 1
Variance predictors	192 channels; kernel size 3; dropout 0.3
Pitch and energy embeddings	256 bins for each feature
Mel projection and postnet	80 mel channels; five convolutional layers; 256 hidden channels

The variance adaptor has three predictors: duration, pitch and energy. Training uses targets extracted from the recordings; synthesis uses the model's predictions. The length regulator repeats each token's hidden features for the predicted number of frames. The decoder then produces a mel spectrogram, which the postnet refines.

This implementation uses the reference code's token-level pitch option: it predicts mean log F0 for each token rather than the continuous wavelet representation in the original paper. The tokens are romanized characters, not phonemes. The code still calls this feature level phoneme_level, so that name should not be read as a phoneme-based frontend. Romanization lets the model and aligner use the same text, but loses some pronunciation distinctions.

The code changes also fix the energy-control path, keep length-regulator tensors on the correct device and prevent long decoder outputs from being truncated. Attention uses PyTorch scaled-dot-product attention. The patch and source-file hashes are saved in docs/upstream.patch and docs/upstream_manifest.json.

SpeechBrain's pretrained multispeaker HiFi-GAN [7] converts the mel spectrograms to 22.05 kHz audio. Its weights are not updated. The loader translates convolution-wrapper parameter names and then checks that every weight matches.

2.1 Acoustic features and objective

Mel spectrograms use a 1,024-sample FFT and window, a 256-sample hop and 80 Slaney mel filters from 0 to 8,000 Hz. Reflect padding and uncentered STFT frames match the vocoder's feature settings. WORLD DIO and StoneMask estimate F0. Unvoiced intervals are filled by interpolation in log F0, then averaged over each token. Energy is calculated from the L2 norm of the STFT magnitude and averaged in log space over the same intervals.

Only training tokens are used to calculate pitch and energy normalization statistics and quantization bounds. Validation and synthesis reuse these values. The total loss adds mel and postnet-mel L1 losses to MSE losses for log(1 + duration), normalized log F0 and normalized log energy. Padding is masked throughout; tokens with zero duration are also excluded from pitch and energy loss.

3. Dataset preparation

The English recordings come from Svarah [3], which contains Indian-accented English speech. The Telugu recordings come from SPRINGLab's IndicTTS Telugu release [4]. Dataset revisions are recorded in the configuration files. The Telugu release contains two voice labels, supplied through its gender field.

Svarah does not provide speaker IDs in this release. For English speaker conditioning, a hash is made from eight profile fields: gender, age group, primary language, native state, native district, qualification, job category and occupation domain. This gives 115 groups in the accepted data. A group is not a verified speaker: people with the same profile share an embedding. Telugu uses the two supplied voice labels.

Preprocessing removes silent or invalid recordings, clips shorter than 0.7 seconds or longer than 20 seconds, and transcripts longer than 450 tokens. Audio hashes identify duplicate recordings. Normalized transcripts are grouped before splitting so the same prompt cannot appear in both training and validation. The split also reserves training examples for each conditioning group. Voices or profile groups may appear in both splits; recordings and normalized prompts do not overlap.

The candidate pool includes 15% extra audio to allow for failed alignments. The five-hour training and one-hour validation targets are checked again using the accepted recordings. Split seeds are 20261101 for English and 20261102 for Telugu.

Text is normalized to Unicode NFC, and rules expand numbers, dates, ordinals and times. Telugu is romanized with uroman [6]. Alignment, training and inference use this same frontend. Original text is saved alongside the normalized form. The unusual internal spacing in E3 and T3 is retained because it is part of the test.

Audio is converted to mono at 22,050 Hz. MMS forced alignment [5] estimates character boundaries. Gaps between words and at the start or end of a recording are assigned to boundary tokens. These timings become integer frame counts whose sum must equal the mel-spectrogram length.

4. Training configuration

Table 2. Optimization settings for each language.

Parameter	Value
Optimizer	AdamW; betas (0.9, 0.98); epsilon 1e-9
Peak learning rate and schedule	0.0005; 1,000-step warmup, then inverse-square-root decay
Batch size	6 training; 4 validation
Weight decay and gradient clipping	1e-6 and 1.0, respectively
Validation interval and maximum steps	1,000 and 40,000, respectively
Early stopping	After 15,000 steps; 12 checks without a 0.002 improvement

At each validation check, the model processes the full validation split using ground-truth duration, pitch and energy. Losses are averaged over valid elements, and their sum is used to select the checkpoint. Test synthesis uses predicted duration, pitch and energy, so validation loss alone does not measure final speech quality. Training uses CUDA mixed precision, with BF16 where supported and FP16 otherwise.

Accepted data after preprocessing and alignment

Language	Training utterances	Validation utterances	Conditioning groups
English	3,220	655	115
Telugu	1,922	382	2

5. Experimental results

5.1 English results

Run ended	Selected checkpoint	Selected validation loss
19,000 steps	Step 7,000	2.7096

Validation loss fell from 3.5498 at the first check to 2.7359 at the final check. Its lowest value was 2.7096 at step 7,000. The run ended after 12 checks without an improvement large enough to replace the saved checkpoint.

The checkpoint from step 7,000 was used for all five test utterances.

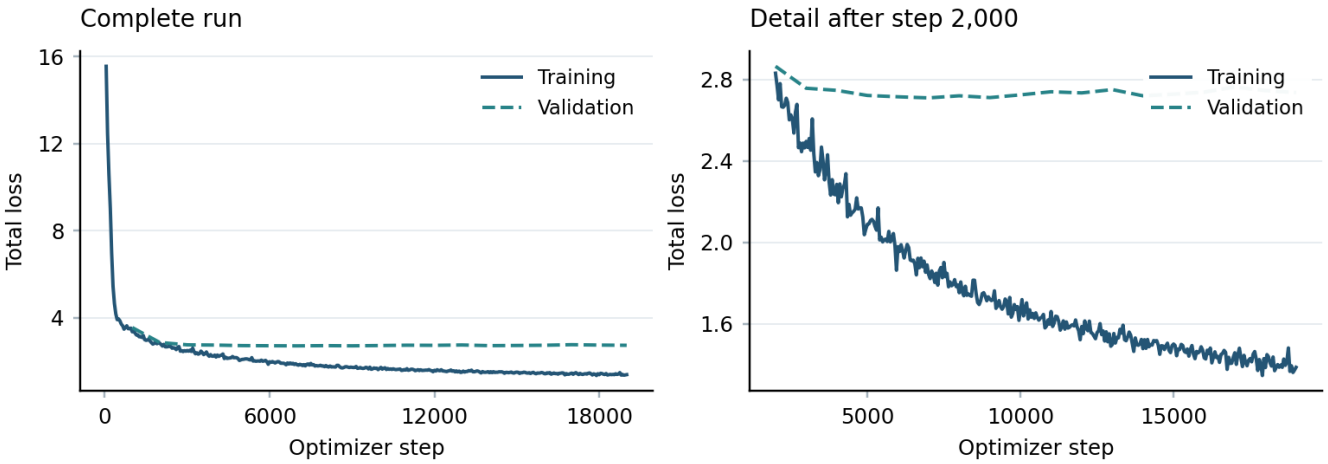


Figure 1. Training and validation loss. The right panel enlarges the later part of the run. Separate loss-component plots are saved in Results/English.

Table 3. Test-sentence measurements for English.

Sample	Duration (s)	Predicted F0 range (Hz)	Measured F0 median (Hz)	Alignment confidence	Zero-duration tokens
E1	6.34	114-173	120.9	0.346	0
E2a	5.19	97-182	109.2	0.329	0
E2b	4.27	84-165	119.0	0.313	0
E3	2.96	117-173	120.8	0.204	0
E4	6.78	105-171	117.5	0.326	2

All five generated samples had lower alignment confidence (0.204-0.346) than the mean for ten held-out recordings (0.746). The gap is a reason to review the English audio closely despite the falling training loss. E3 scored lowest, but the score does not identify a particular word or pronunciation error.

Alignment confidence is the MMS CTC aligner's score for matching the supplied transcript to the audio. The held-out recordings provide a baseline for the same check. This uncalibrated score is not pronunciation accuracy or a listener rating.

No clipped samples were found in the five waveforms. The F0 columns describe different measurements: the predicted range covers the continuous token contour, while the measured median uses voiced regions of the generated audio.

5.2 Telugu results

Run ended	Selected checkpoint	Selected validation loss
26,000 steps	Step 14,000	2.0241

Validation loss fell from 2.5752 at the first check to 2.0433 at the final check. Its lowest value was 2.0231 at step 23,000. The run ended after 12 checks without an improvement large enough to replace the saved checkpoint.

The step-23,000 loss was slightly lower than the saved checkpoint's loss, but the difference was below the 0.002 replacement threshold. Test audio therefore uses the checkpoint from step 14,000.

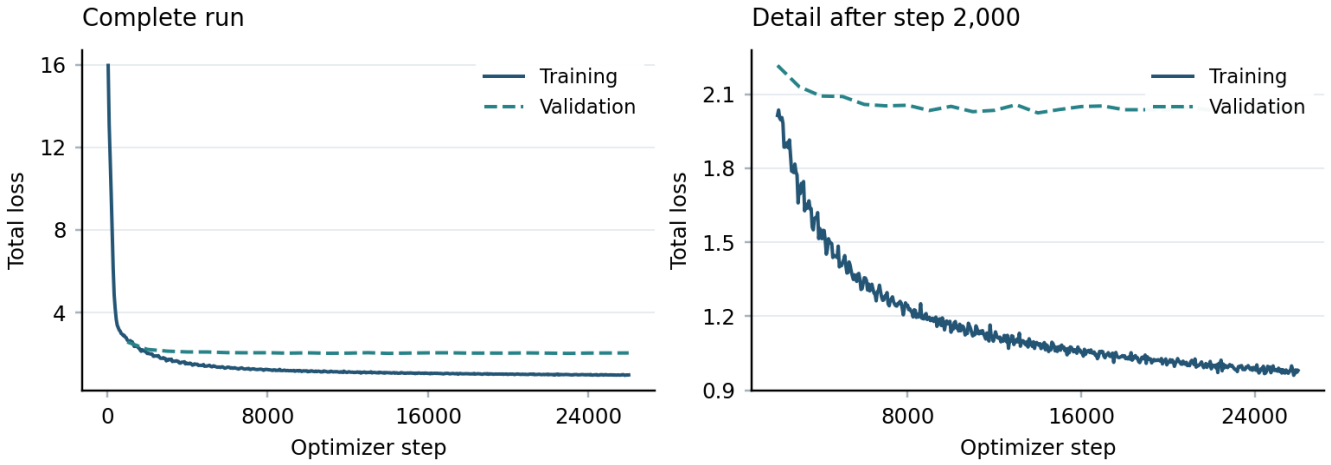


Figure 2. Training and validation loss. The right panel enlarges the later part of the run. Separate loss-component plots are saved in Results/Telugu.

Table 4. Test-sentence measurements for Telugu.

Sample	Duration (s)	Predicted F0 range (Hz)	Measured F0 median (Hz)	Alignment confidence	Zero-duration tokens
T1	6.05	100-294	199.7	0.876	0
T2a	3.33	113-247	185.1	0.980	0
T2b	3.54	89-252	178.8	0.918	0
T3	3.23	106-270	173.2	0.863	0
T4	9.83	100-258	185.2	0.812	1

Generated-sample alignment confidence ranged from 0.812 to 0.980; the mean for ten held-out recordings was 0.946. T2a scored highest and T4 lowest. These values support a more favorable alignment check than the English run, but they do not establish correct stress, phrasing or pronunciation.

No clipped samples were found in the five waveforms. The F0 columns describe different measurements: the predicted range covers the continuous token contour, while the measured median uses voiced regions of the generated audio.

6. Evaluation of the test sentences

Each model generated five utterances, counting the alternatives in E2 and T2 separately. Table 5 lists what needs to be checked in each pair. Listening notes have not yet been completed, so these are evaluation criteria rather than claims that the models passed the tests.

Table 5. Sentence-level evaluation criteria.

Samples	Aspect examined
E1 and T1	Pronunciation and completeness of the rare English word and Telugu technical loanword
E2a and E2b	Context-dependent stress in the noun and verb uses of conduct
T2a and T2b	Phrase-level prosody for the two uses of pattu; the meaning difference need not alter its segmental pronunciation
E3 and T3	Effect of the supplied internal word spacing on segmentation and timing
E4 and T4	Date and time verbalization; E4 also contains the romanized Telugu word andaroo

Each utterance has three plots: token duration in milliseconds, continuous F0 in Hz and energy in relative STFT-magnitude units. The numerical traces are saved as CSV and NPZ files. Since F0 is interpolated through unvoiced regions, a pitch value over a consonant or silence does not mean that region is voiced. The plots show the predicted contours; listening is needed to judge stress and pronunciation.

7. Limitations

The frontend uses romanized characters rather than language-specific phonemes. It can therefore merge sounds that should remain distinct. Errors in forced alignment can also give the duration predictor incorrect targets. Continuous F0 does not explicitly model voiced and unvoiced regions, and the fixed vocoder may not suit every voice or recording condition.

The English and Telugu losses should not be used to rank the two models: the datasets, voices and normalization statistics differ. There are no reference recordings of the test sentences for a direct audio comparison. A useful next check is to pass held-out ground-truth mel spectrograms through the vocoder. Comparing those outputs with model-generated speech would help identify which errors come from the acoustic model and which come from the vocoder.

8. Reproducibility

The language configuration files record dataset revisions, feature settings, model dimensions, seeds and optimizer settings. Each checkpoint includes the configuration, token inventory, speaker mapping and training statistics. Separate files save optimizer and random-number states for resuming training. Dependencies are listed in requirements.txt, and the README gives the setup and commands for each stage. The full pipeline can be run with:

```
python scripts/run_project.py --work-dir PATH --wait-for-access
```

Fixed seeds make the split repeatable, though training results can still vary across CUDA hardware and software versions. Raw recordings and access credentials are kept outside the repository.

References

[1] Y. Ren et al., "FastSpeech 2: Fast and High-Quality End-to-End Text to Speech," ICLR, 2021. [online]

[2] ming024, FastSpeech2 reference implementation, commit d4e79eb52e8b01d24703b2dfc0385544092958f3. [online]

[3] T. Javed et al., "Svarah: Evaluating English ASR Systems on Indian Accents," Interspeech, 2023, pp. 5087-5091. Dataset: [online]

[4] SPRINGLab, IndicTTS Telugu dataset release, derived from the IIT Madras Indic TTS database. [online]

[5] TorchAudio 2.6, "Forced alignment for multilingual data," MMS_FA documentation. [online]

[6] U. Hermjakob, J. May and K. Knight, "Out-of-the-box Universal Romanization Tool uroman," ACL Demonstrations, 2018. [online]

[7] SpeechBrain, HiFi-GAN trained on LibriTTS at 22,050 Hz, model revision 4188503131602dc234f48d7f22eebea93d788736. [online]